Information Diffusion in Social Network

(SNA – B Tech III CSE)

(Week#4 – Jan 28, 2026)

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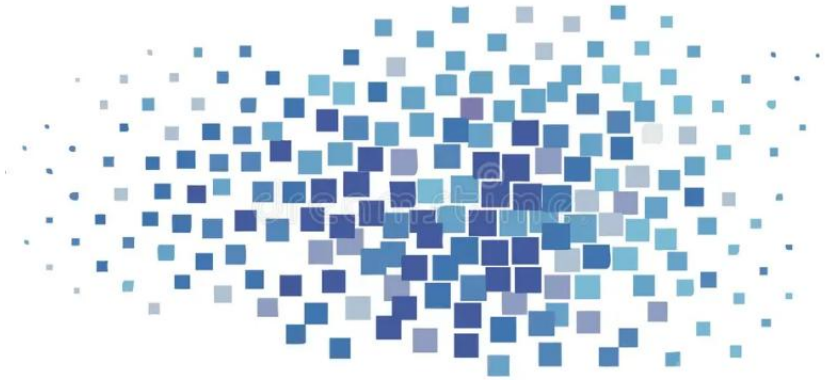


- Disclaimer – subjective opinion

SNA is more of a decision science than arithmetic

- Gold price moved from INR 80k/g to INR 163k/g in last one year
- Silver price moved from INR 75k/kg to INR 325k/kg (av. Production cost is just around INR 65k/kg) – Banks are selling paper silver (shorts)
- Store of Value? Proven Hedge? Fiat Currency is weakening day by day… US govt. shutdown in the next 4-5 days

Silver: Likely vulnerable to correction because price has detached from cost and supply

- Silver: Fundamentals do **not** justify parabolic pricing
- Production cost has not risen proportionally
- Geological scarcity is not the binding constraint (reserves depleting?)
- Silver is a **hybrid asset**:
- Part precious metal, part industrial commodity, part speculative instrument.
- That makes pullbacks not only likely, but structurally normal after vertical moves.

Gold responds to **loss of credibility**, not loss of supply

- Gold's strength is not about metal scarcity. It is about **institutional distrust in monetary governance**.
- Gold pricing is driven primarily by: Monetary trust (or distrust), Sovereign risk perception, Central bank balance sheets, Real interest rates, Geopolitical instability
- It is **not** priced on: Industrial utility, Production cost curves, Consumption demand
- Geopolitical fragmentation of reserve systems and De-dollarization at the margin -

What would USD cracking actually look like?

- If this is unfolding in real time, we would see:

1. US Treasury auctions failing
2. Long-term yields spiking uncontrollably
3. Dollar collapsing across all currency pairs (not selectively)
4. Capital controls discussion inside the US
5. Emergency IMF interventions
6. Rapid reserve dumping by major allies (Japan, EU)
7. Domestic banking instability

Nodes (actors / variables / constructs)

- These are not people; they are **structural concepts** in a system.
- Examples from our discussion: These are **conceptual nodes** in a political–economic system.

Nodes	
USD credibility	Institutional stability
Gold price	Capital flows
Silver price	Central bank behavior
De-dollarization	Sanctions / tariffs
Treasury auctions	Power projection
Bond yields	Fiat debasement
Elite cohesion	Market volatility
Public trust	System legitimacy
Leadership legitimacy	

Edges = causal, reinforcing, or inhibitory relationships

From Node	To Node	Edge Type
Fiat debasement	Gold price	Positive reinforcement
Elite cohesion	System stability	Positive reinforcement
Leadership decline	Public trust	Negative impact
Public trust	USD credibility	Positive reinforcement
De-dollarization	Gold demand	Positive reinforcement
Treasury auction stress	Bond yields	Positive feedback
Sanctions overuse	USD legitimacy	Negative impact
Paper silver leverage	Silver volatility	Positive amplification
Market volatility	Demand for gold	Positive reinforcement

Clusters (communities of concepts)

- We will see natural clusters like:

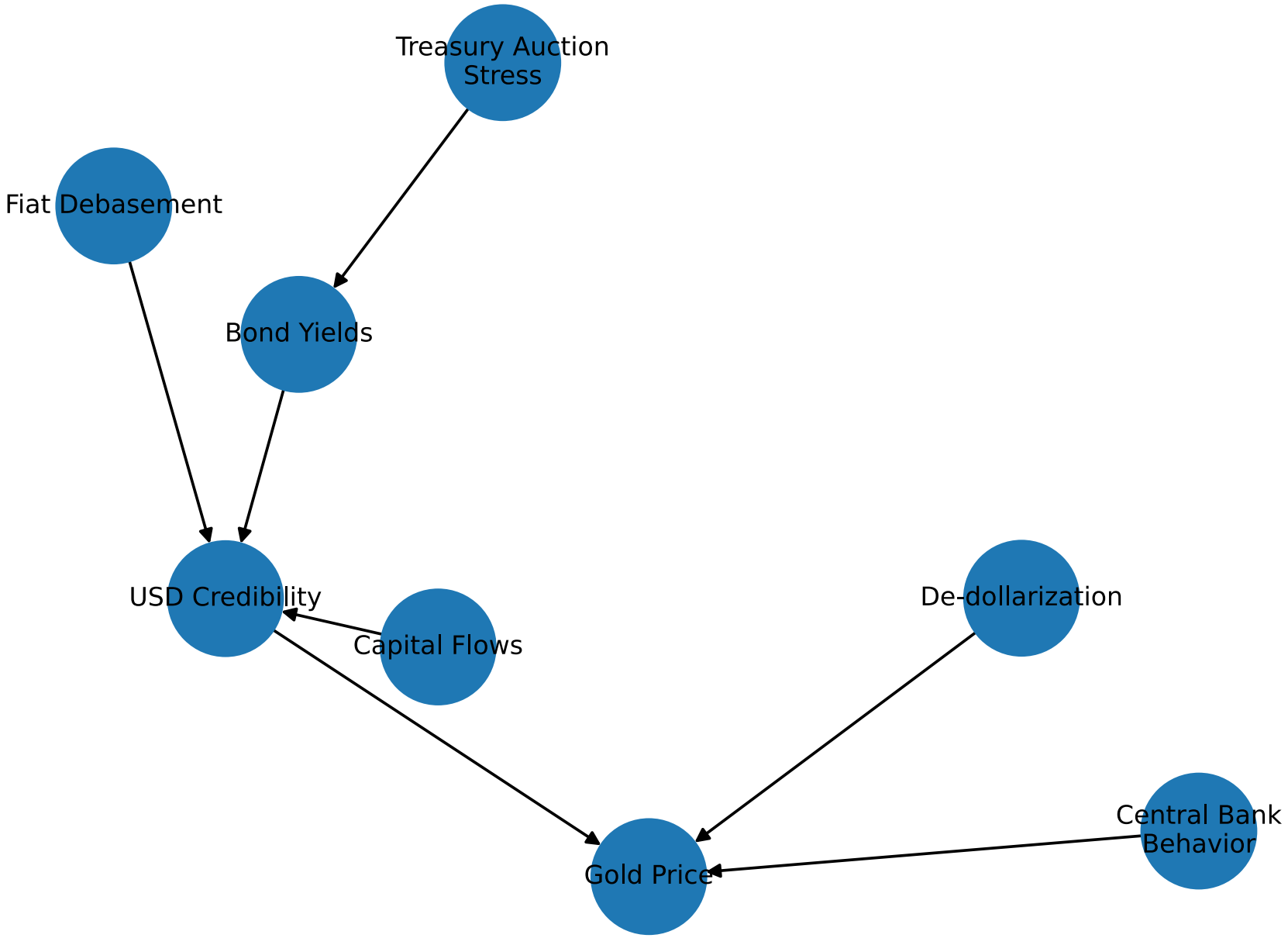
Monetary cluster USD credibility Fiat debasement Gold price Bond yields Inflation expectations	**Political legitimacy cluster** Leadership credibility Public trust Institutional stability Elite cohesion	**Market mechanics cluster** Paper silver leverage Futures positioning Physical supply Volatility

Edges = typed relationships

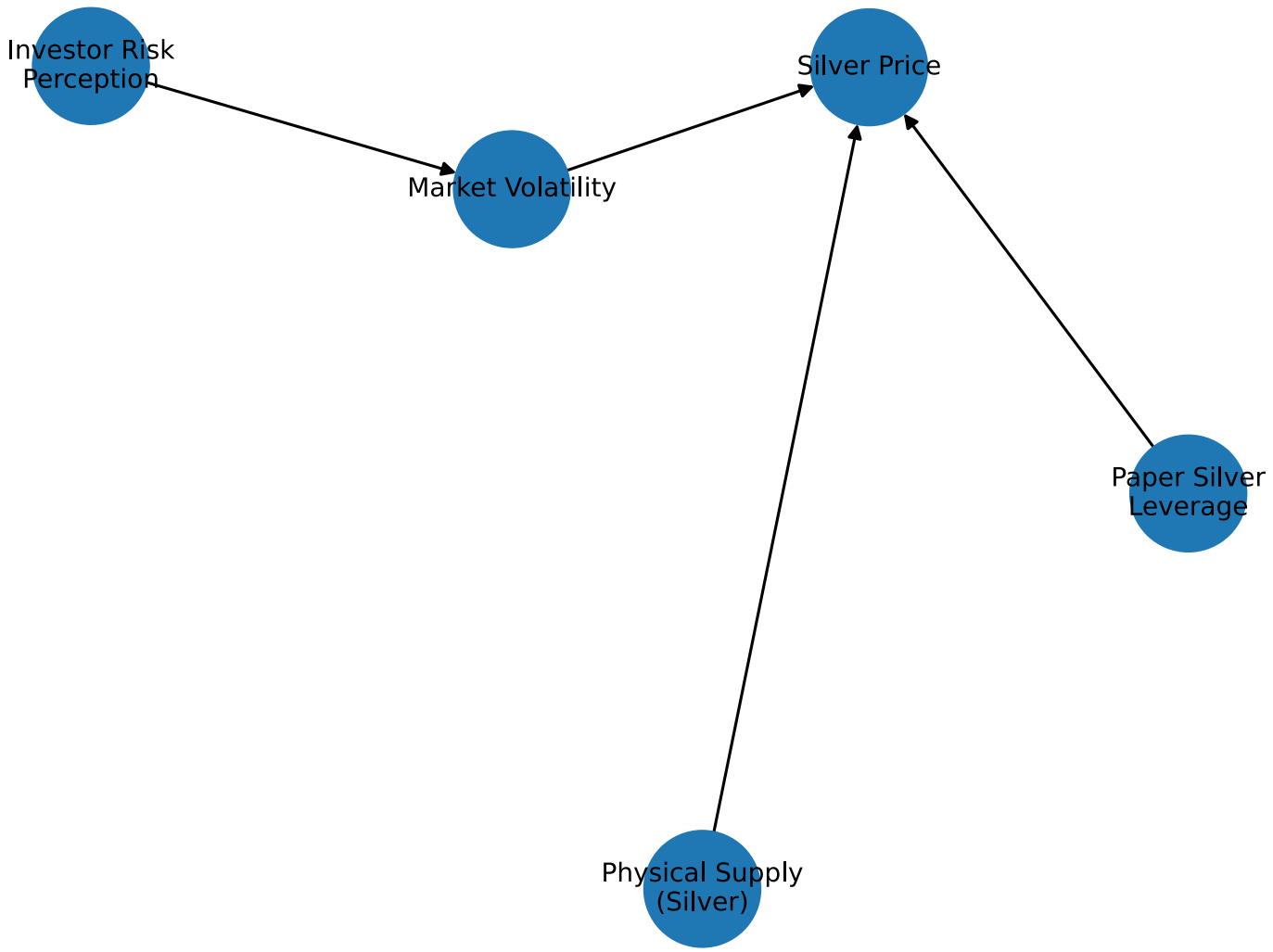
- Each edge has: Source, Target and Relationship (Positive / Negative)

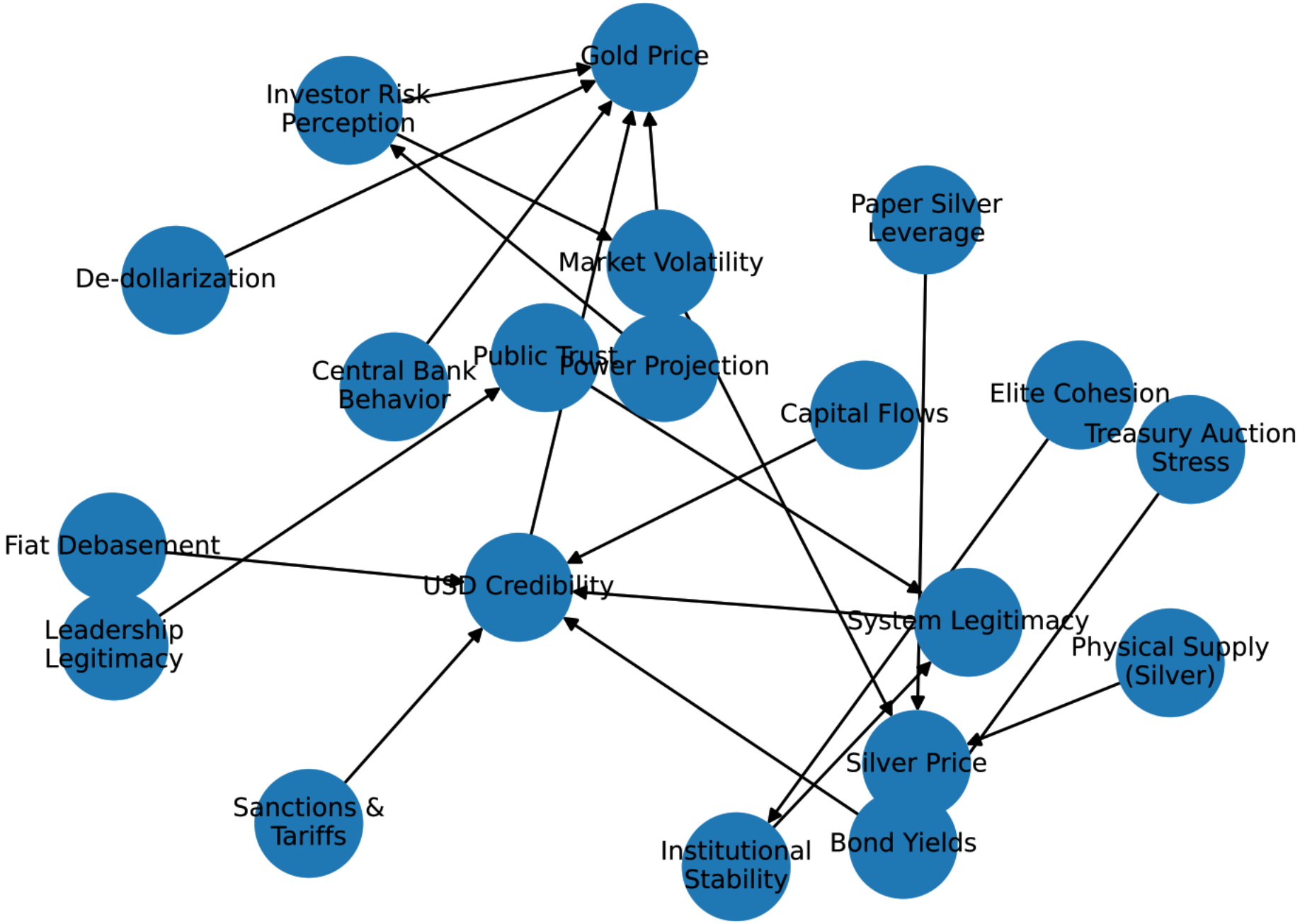
Source	Target	Meaning
Fiat Debasement → USD Credibility	Negative	More debasement erodes trust
Leadership Legitimacy → Public Trust	Positive	Better leadership increases trust
Public Trust → System Legitimacy	Positive	Trust stabilizes system
System Legitimacy → USD Credibility	Positive	Legitimacy supports currency
De-dollarization → Gold Price	Positive	Drives gold demand
Paper Silver Leverage → Silver Price	Positive	Amplifies volatility upward
Physical Supply (Silver) → Silver Price	Negative	Supply caps price
Treasury Auction Stress → Bond Yields	Positive	Stress pushes yields higher

Gold price


Treasury Auction Stress
Fiat Debasement
Bond Yields
USD Credibility
Capital Flows
De-dollarization
Central Bank Behavior
Gold Price

Silver price




Gold Price
Investor Risk Perception
Paper Silver Leverage
Market Volatility
De-dollarization
Public Trust
Power Projection
Central Bank Behavior
Elite Cohesion
Capital Flows
Treasury Auction Stress
Fiat Debasement
USD Credibility
System Legitimacy
Leadership Legitimacy
Physical Supply (Silver)
Silver Price
Sanctions & Tariffs
Institutional Stability
Bond Yields

Don’t worry much →

🇮🇳 Indian women hold more Gold than 10 countries combined:

🇮🇳 Indian Women: 25,488 tonnes

1. 🇺🇸 USA: 8,133 tonnes
2. 🇩🇪 Germany: 3,351 tonnes
3. 🇮🇹 Italy: 2,451 tonnes
4. 🇫🇷 France: 2,437 tonnes
5. 🇷🇺 Russia: 2,332 tonnes
6. 🇨🇳 China: 2,279 tonnes
7. 🇨🇭 Switzerland: 1,039 tonnes
8. 🇯🇵 Japan: 845 tonnes
9. 🇳🇱 Netherlands: 612 tonnes
10. 🇵🇱 Poland: 448 tonnes $XAU

Discussion on Examination during Covid-time

- Non-proctored mode
- Non googleable/bingeable MCQs
- 50 different question papers of same subject (course) for a class
- Time squeeze, Interactive Exam Control Room Dashboard – bringing students to exam on time.
- Coursera adoption with 4500 licenses (with 6 month validity) given to students
- Faculty members asked to align respective course to at least 5 related courses on Coursera
- Wall of certificates
- Aligning students to appear for exam (degree certificate as they complete on schedule#1)

Back to the planned business - What?

- Why do some tweets go viral and others die instantly?
- How did misinformation spread during COVID-19?
- Why do some products (e.g., iPhone) diffuse faster than others?
- How do political narratives propagate through societies?
- Information does not spread randomly. It spreads through structure.
- Diffusion is a **network process**, not just an individual behavior.
- **Diffusion is not about content alone — it is about structure**

Case Study

- Early COVID-19 vaccine resistance in parts of Europe emerged not because of ignorance, but because of **how trust, identity, institutions, and information networks are structured**.
- Disclaimer - This is a conceptual diffusion illustration

Why?

- In several European countries early in the rollout:
- Trust in:
  - Government
  - Pharmaceutical companies
  - EU-level institutions
  - Media - was already weakened (pre-existing political and social context).
- So even when information flowed, it arrived through **low-trust edges**.
- The edge existed (information reached people) but its **weight was near zero**.

Why?

- In Europe there was:
- AstraZeneca pauses and reversals
- Conflicting national guidelines
- Age restrictions changing week to week
- Public disagreement among experts
- From a diffusion perspective:
- Instead of reinforcing belief, exposure reinforced **ambiguity**.
- This produced:
- Cognitive hesitation
- Threshold not crossed
- Adoption cascades stalled

Why?

- Vaccine resistance did not emerge uniformly.
- It clustered along:
- Political identity
- Cultural identity
- Historical attitudes toward authority
- Regional networks
- Through homophily:
- Skeptical individuals were more connected to skeptical individuals
- Resistant narratives circulated more densely inside those clusters
- That created **localized echo chambers** of resistance.

Why?

- Many European societies have:
- Strong civil-liberty traditions
- Deep historical sensitivity to state coercion
- Strong individual autonomy norms
- So when vaccination began to be framed (in some countries) as:
- Mandates, Passports, Restrictions
- The diffusion mechanism flipped:
- Information exposure did not increase adoption, it triggered **reactance**.
- In threshold models, this looks like:
- Threshold increases instead of decreases when pressure rises.

Why?

- **Social media accelerated polarized diffusion, not neutral diffusion**
- Platforms did not merely transmit information. They:
- Amplified emotionally charged content
- Prioritized engagement
- Strengthened cluster boundaries
- This meant:
- Pro-vaccine messages spread rapidly within some communities
- Anti-vaccine narratives spread rapidly within others
- Very little effective cross-community diffusion
- Structurally:
- High internal diffusion + low inter-community bridging = polarization.

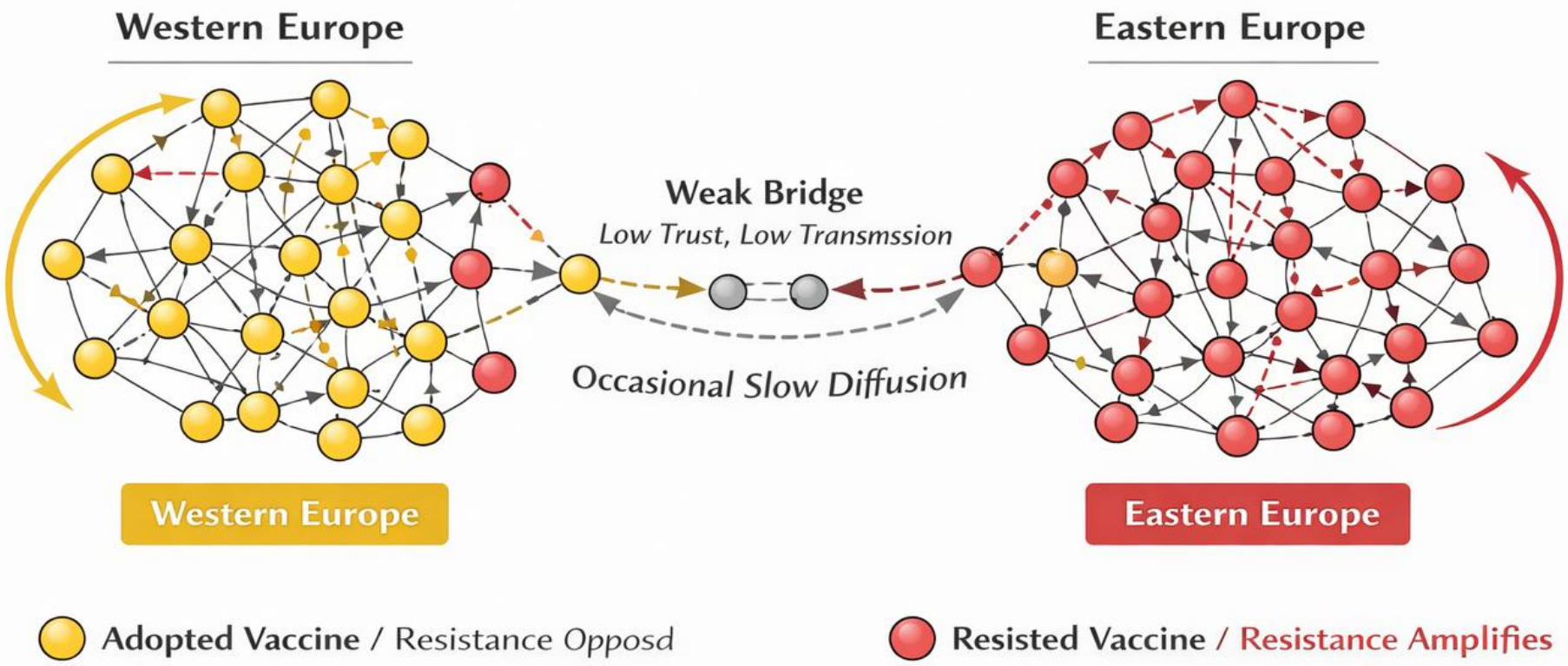
Why?

- Compared to some other regions:
- Europe had:
  - High digital connectivity
  - Strong activist networks
  - Strong civil rights discourse
  - Politically fragmented trust environments
- Which meant:
  - Diffusion of skepticism was **fast**
  - Institutional correction lagged behind network dynamics
- Once a belief cluster stabilizes early, later corrections struggle to penetrate

Information Diffusion

COVID-19 Vaccine Resistance in Europe

Resistance Amplifies Within Groups, Blocks Across Groups



Summary

Concept	COVID case
Homophily	Skeptic clusters formed
Echo chambers	Reinforcement of resistance
Threshold models	People didn’t cross adoption threshold
Weak ties	Few bridges between pro- and anti-groups
Cascade failure	Adoption stalled in some communities
Reactance	Pressure increased resistance
Network trust	Source credibility mattered more than facts

Catalyst

- **Novak Djokovic** — the Serbian world-number-one **did not receive a COVID-19 vaccine and therefore missed several major tournaments** when vaccine mandates were in place
- He was **denied entry into Australia in early 2022 and subsequently deported** because he was unvaccinated and could not satisfy the country's vaccine requirements at the time.

Further Analysis

- **He opposed mandates, not necessarily vaccines themselves**
- He repeatedly emphasized:
- He was **not anti-vaccine in general**, but opposed to being **forced to take a medical intervention** as a condition to work or travel.
- His core claim:
- “The principle of making a decision over my own body is more important than any title or tournament.”

Further Analysis cont.

- **He framed it as a matter of bodily autonomy**
- He described his stance as:
- A matter of **individual sovereignty**
- A matter of **freedom of choice**
- A concern about **coercion by institutions**
- This is why his stance resonated strongly with:
- Civil-liberty-oriented audiences
- Anti-mandate communities
- People already skeptical of authority
- From a diffusion perspective, this made his message an **identity signal**, not a medical argument.

Analytically interesting case

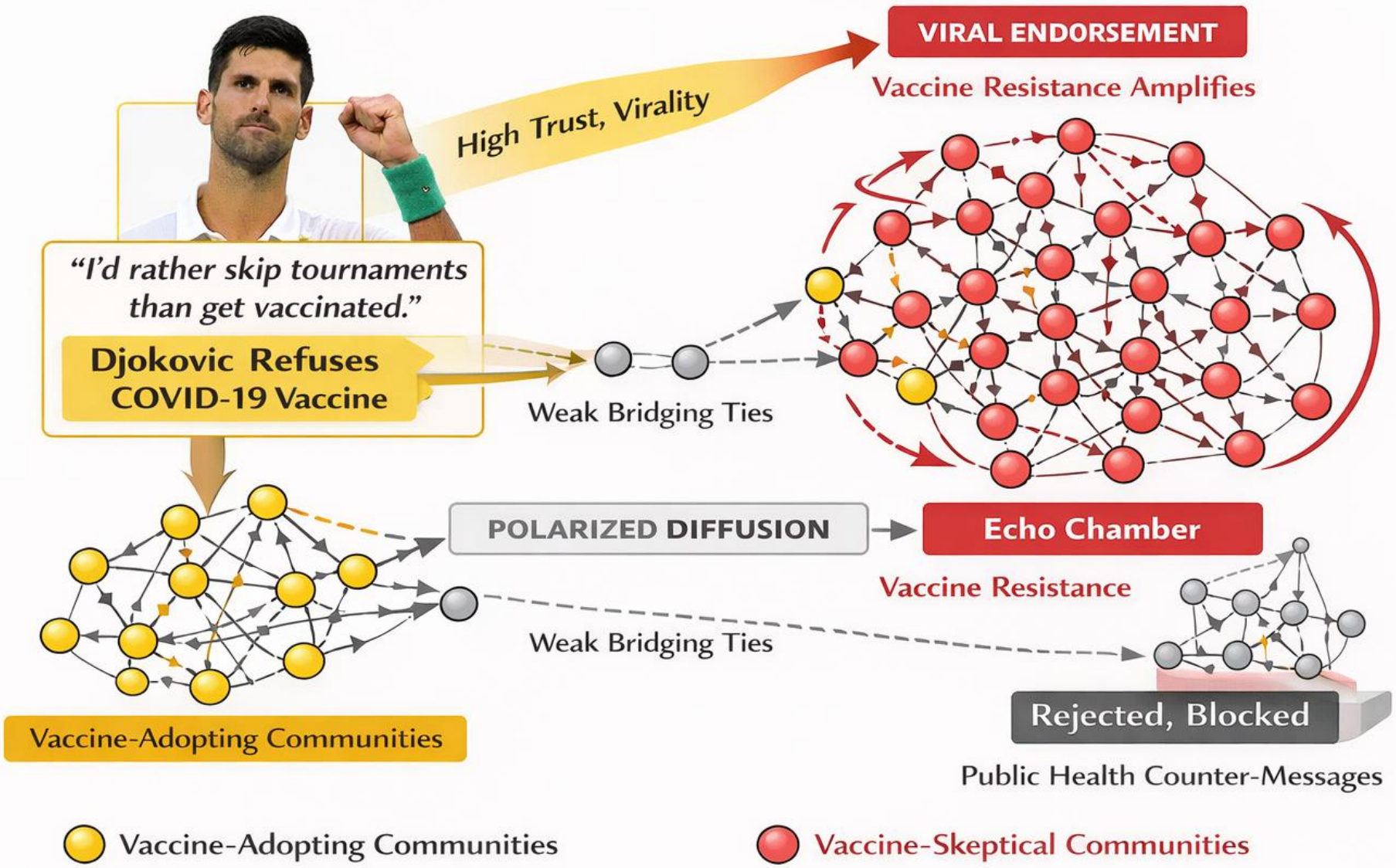
- **He avoided making medical claims**
- He did **not** publicly claim vaccines are harmful
- He did **not** promote alternative treatments
- He did **not** present himself as a medical authority
- The diffusion was driven by **symbolism**, not by technical misinformation.

Djokovic case - summary

Concept	Case Illustration
High centrality node	Celebrity athlete
Seed selection	Initial diffusion point
Homophily	Belief clusters among supporters
Echo chamber	Reinforcement within group
Social proof	“If he believes it, it must be valid”
Prestige bias	High-status individual influence
Weak ties failure	Counter-message couldn’t penetrate
Cascade	Viral endorsement spread rapidly

Novak Djokovic and Vaccine Resistance

Celebrity Endorsement as a Diffusion Cascade



Summary

- The message:
- Crossed into political identity networks
- Triggered emotional resonance
- Activated existing distrust in institutions
- Spread via **prestige bias + identity alignment**
- Djokovic's message was not 'vaccines are dangerous,' but 'no one should be forced to decide for my body.'
- The diffusion occurred because this message aligned with identity and trust structures in certain communities
- The most viral messages are not those with the most evidence, but those that best align with identity, trust, and network position.

Takeaway

- Networks shape belief
- Structure beats content
- Position beats persuasion
- Trust beats information